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A preliminary Upper Triassic paleomagnetic pole for the Khorat plateau (Thailand): consequences for the accretion of Indochina against Eurasia

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We report new paleomagnetic results from the Upper Triassic (Norian) Huai Hin Lat Formation at the base of the Khorat Group in Central Thailand. Five sites of conglomerate, sandstone and limestone reveal a stable component of magnetization ($D = 43°$, $I = 43°$; $\alpha_{95} = 7.5°$). We argue that this component is of primary origin on the basis of a positive fold test, observation of both normal and reversed directions and a reported magnetostratigraphy from Middle Triassic to Cretaceous. The new Triassic pole yields, together with published younger poles, a Mesozoic apparent polar wander path for Indochina. This unit is found to have remained at tropical northern latitudes since the Upper Triassic. Comparison with Eurasian poles reveals 1650 ± 850 km of SW-NE convergence and $15° \pm 10°$ of counterclockwise rotation between 205 and 160 Ma. Two mechanisms are discussed to account for this motion. The first one implies the closure of an oceanic basin to the north of the South China block. The second one involves post-welding intracontinental deformation.

1. Introduction

South-East Asia is now recognized as a composite mosaic which grew to its present size through the repeated accretion of continental fragments throughout the Upper Paleozoic and the Mesozoic [1,2]. This mosaic was extensively reshaped and redistributed as a consequence of the Cenozoic collision of India against Asia. Continental blocks with a Paleozoic core that have been recognized in southeastern Asia include the Shan-Thai (Burma, West Thailand, West Malaya), Indochina (or Annamia), South China, Southern Tibet and Northern Tibet (Fig. 1) [2–7]. These fragments are sometimes traced back to a Cimmerian [8] or Pacifica [9] continent, of uncertain Gondwanian origin.

The extent to which Indochina was affected by the India-Eurasia collision was suggested by Molnar and Tapponnier [10] and Tapponnier and Molnar [11,12] and emphasized by Peltzer et al. [13] and Tapponnier et al. [7,14]. The southeastward extrusion of Indochina envisioned by these authors can be tested by paleomagnetic studies. A mostly successful first-order test was made by Achache et al. [15] on the basis of paleomagnetic data available at the time, for the Cretaceous and Cenozoic. Many additional data have been made available by Maranate [16] and Maranate and Vella [17] from the Lower Jurassic of the Khorat plateau upwards. This data set constrains the large clockwise rotation of the Indochina block with respect to Eurasia to being later than Middle Cretaceous. The Upper Jurassic and Lower Cretaceous are, on the contrary, geological periods where no significant motion of Indochina with respect to Asia is observed [16].

Almost no reliable paleomagnetic data were available for pre-Jurassic time in the Indochina

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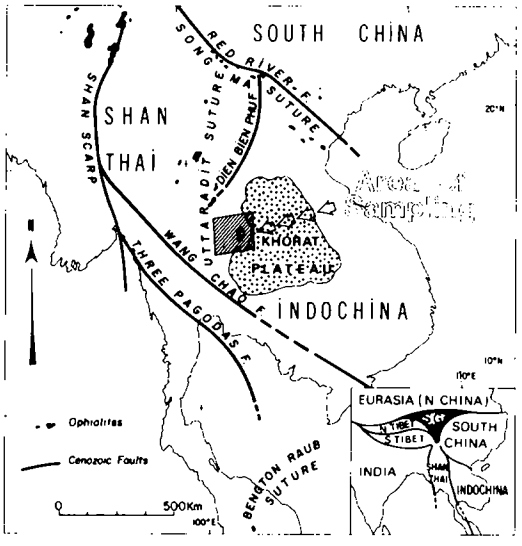


Fig. 1. Schematic tectonic map of South-East Asia showing the upper Paleozoic to lower Mesozoic suture zones (ophiolites) and the main Cenozoic faults (after Tapponnier et al. [7]). Also indicated is the area of sampling. The insert shows several independent continental blocks recognized in South-East Asia. *S.G.* = Songpan-Ganzi or Bakan Hai range.

block. This is the time when Indochina is generally thought to have been accreted to South China along the Song Ma suture in the Lower Triassic and to the Shan-Thai block along the Uttaradit suture in pre-Rhaetian times (Fig. 1) [2,6,7,16,18]. This paper reports the results of a preliminary paleomagnetic study of Upper Triassic (Norian) sedimentary rocks at the base of the mildly folded Khorat Group. The new Triassic pole can be integrated with younger poles [16,19,20], yielding an apparent polar wander path for this block in the Mesozoic. This APWP places constraints on the process of accretion of Indochina against the southeastern margin of Asia.

2. Age and tectonic setting

The Khorat Group (Fig. 1) consists of a thick series (4000 m) of mostly continental Mesozoic sediments ranging from Upper Triassic to lower Cenozoic in age (e.g. [21]). This series is only mildly folded, with dips usually on the order of 5 or 10°, except for its western margin, where dips may locally reach 25°. It lies unconformably over folded Paleozoic formations which are exposed on the edges of the plateau. The youngest of these folded sediments are middle Permian Fusulina limestones which appear as rugged cliffs, forming one of the characteristic landscapes of southeastern Asia.

The Khorat Group is divided into several formations. The lowermost member of the group is the Huai Hin Lat Formation, characterized by a basal conglomerate containing pebbles of the underlying Permian limestones, overlain by predominantly continental and lacustrine deposits, including intercalated sandstones, shales and limestones. Joint Thai-French paleontological studies have provided both stratigraphic and paleogeographic control on the Khorat Group (see, e.g., Buffetaut [22,23]). In particular, the Huai Hin Lat Formation has been given a precise Upper Triassic Norian age on the basis of vertebrate fossils [24–28].

The occurrence of the continental Khorat Group lying unconformably over neritic middle Permian limestones, and of a sedimentary hiatus between the middle Permian and Upper Triassic, is generally thought to correspond to the closure of an oceanic domain in Central Thailand and suturing of the Shan-Thai block with Indochina sometimes prior to the Norian. Volcanic and volcanodetritic rocks are often observed in Central Thailand near the edge of the Khorat plateau and are considered as remnants of Permo-Triassic volcanism associated with this collision. However, no radiochronological age is available for these rocks, at least in the area surveyed.

The identification of the Triassic suture between the Shan-Thai and Indochina blocks is by no means straightforward [2,4,18]. Two separate strings of ophiolites are apparently found in central Thailand (Fig. 1). However, they may not be remnants of a single suture [18]. The Uttaradit suture does not appear to have been affected by Cenozoic tectonics, contrary to other sutures in southeastern Asia [4,7]. The situation is even more complex along the northeastern boundary of the Indochina block, where the Triassic Song Ma suture zone has been extensively altered by Cenozoic left-lateral and recent right-lateral motion on the Red River fault [7,14]. Tapponnier and his colleagues also

infer left-lateral Cenozoic slip on the Wang Chao and Three Pagodas faults to the southwest of the Khorat (Fig. 1), and conjugate slip on the Dien Bien Phu fault to the northwest, but the amplitude of faulting there may have been significantly less than on the Red River and Shan Scarp faults (some 10^2 km vs. more than 10^3 km).

The continental mass that had probably formed by the Late Triassic reunion of the Shan-Thai, Indochina and South China Blocks, apparently escaped major deformation in the Jurassic and Cretaceous, but was profoundly remodelled in the Cenozoic, following the collision of India [7,14]. The Upper Triassic paleomagnetic results discussed in this paper are relevant to the Khorat plateau and thus to the quasi-stable Indochina block, as long as one remains away from its boundaries, along which significant shear may have occurred. Care must of course be exercised in attempting to extend these results to the neighbouring blocks such as the Shan-Thai.

3. Paleomagnetic study

Sampling. We sampled oriented cores in central Thailand, along the western escarpment of the Khorat plateau (Fig. 2). Due to the paucity of Triassic outcrops suitable for paleomagnetic studies, only five sites could be sampled in the Huai Hin Lat Formation. Sampling was conducted in parallel with a paleontological expedition [22] which provided reliable age control on the sites.

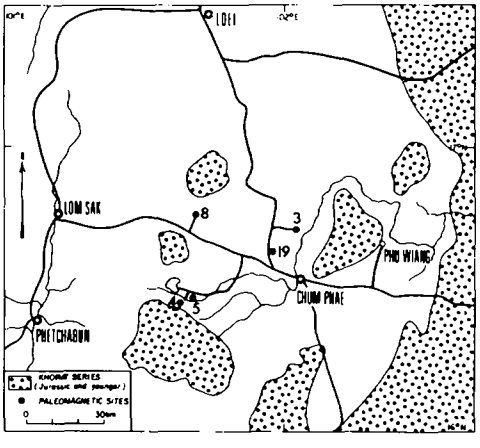


Fig. 2. Detailed geographic map of the area of sampling. Only the Jurassic and younger series of the Khorat have been represented. The rest of the map is covered by Paleozoic marine sediments, Triassic continental deposits and undated volcanics and volcanodetritics, but no detailed geologic map is available for this area.

Site 5 was taken in the basal conglomerate and samples were drilled in the pebbles of Permian Fusulina limestone and in the coarse-grained sandstones matrix. Site 4 is in grey sandstone. Sites 3, 8 and 19 are in limestone. Site 3 is the site of the discovery of a Norian turtle [27], whereas site 8 is a dark bituminous limestone in which many other Norian vertebrate fossils [24] were discovered. Altogether, 39 cores were drilled in the Huai Hin Lat Formation, and 2 or 3 specimens were sawed from these cores and subjected to different cleaning techniques.

Magnetic measurements. Specimens were demagnetized using a laboratory built furnace and a Schönstedt single-axis alternating field demagnetizer. Measurements were performed on a CTF cryogenic magnetometer. The analysis of remanent magnetization proceeded along a classical two-step procedure (cf. [29,30]).

Results. Natural remanent magnetization (NRM) intensities are rather tightly grouped. They range from 10^{-4} to 10^{-3} A/m for sites 3, 4 and 8 and are one order of magnitude stronger in site 19. For the conglomerate (site 5) intensities are between 10^{-3} and 5×10^{-3} A/m in the matrix and between 10^{-2} and 5×10^{-2} A/m in the limestone pebbles. NRM directions are also well grouped and four of them indicate a clockwise rotation with respect to the present field.

Alternating field (AF) and thermal progressive demagnetization revealed two components of magnetization. These two components have strongly overlapping coercivity spectra and thus are difficult to separate by AF demagnetization (Fig. 3). On the contrary, their spectra of blocking temperature are distinct and thermal demagnetization allows a clear identification of both components (Fig. 3). The first component (component A) is destroyed below 300°C, whereas the second component (component B) has a final unblocking temperature of 500–550°C (Fig. 3). These features

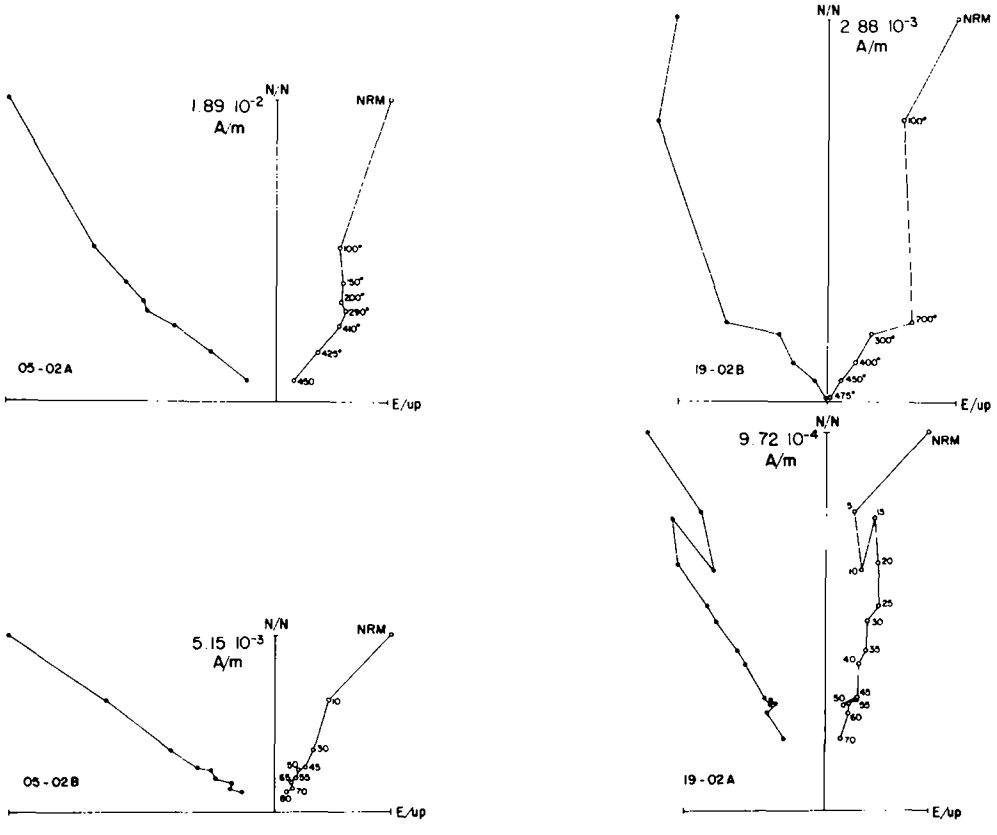
TABLE 1

Paleomagnetic results—Huai Hin Lat Formation

Site	N	Geographic coordinates		Stratigraphic coordinates				VGP	
		$D(°)$	$I(°)$	$D(°)$	$I(°)$	$\alpha_{95}(°)$	K_S	$\lambda(°N)$	$\phi(°E)$
3	5	29.8	41.8	33.2	53.2	5.5	194	55.6	155.9
4	4	45.1	27.0	49.1	37.1	9.4	96	43.4	178.4
5	6	38.9	26.7	44.3	45.4	3.2	451	47.7	169.5
8	4	228.4	−3.5	224.6	−38.3	31.8	9	47.7	177.9
19	7	32.5	38.2	42.2	39.6	4.3	196	49.9	176.6
Mean	5	39.6	27.7	43.2	42.9	7.5	106	49.2	172.3
		$\alpha_{95} = 15.8°$,	$K_G = 24$				$K_S/K_G = 4.4$	$A_{95} = 7.0°$,	$K = 119$

N = number of samples or sites; D, I = declination, inclination; λ, ϕ = coordinates of the virtual geomagnetic poles (VGP); α_{95}, A_{95} = radius of the circle of 95% confidence; K_S, K_G = precision parameters in stratigraphic or geographic coordinates.

point to magnetite or titanomagnetite as the carrier of both components of magnetization. In site 8, component B shows a reversed polarity and required a more sophisticated procedure for separation from the other component. Specimens were first heated to 200 or 300°C and then demagnetized by alternating field up to 80 mT peak values (Fig. 3). This procedure was successful in partially isolating component B in only four specimens. The directions observed may still be su-



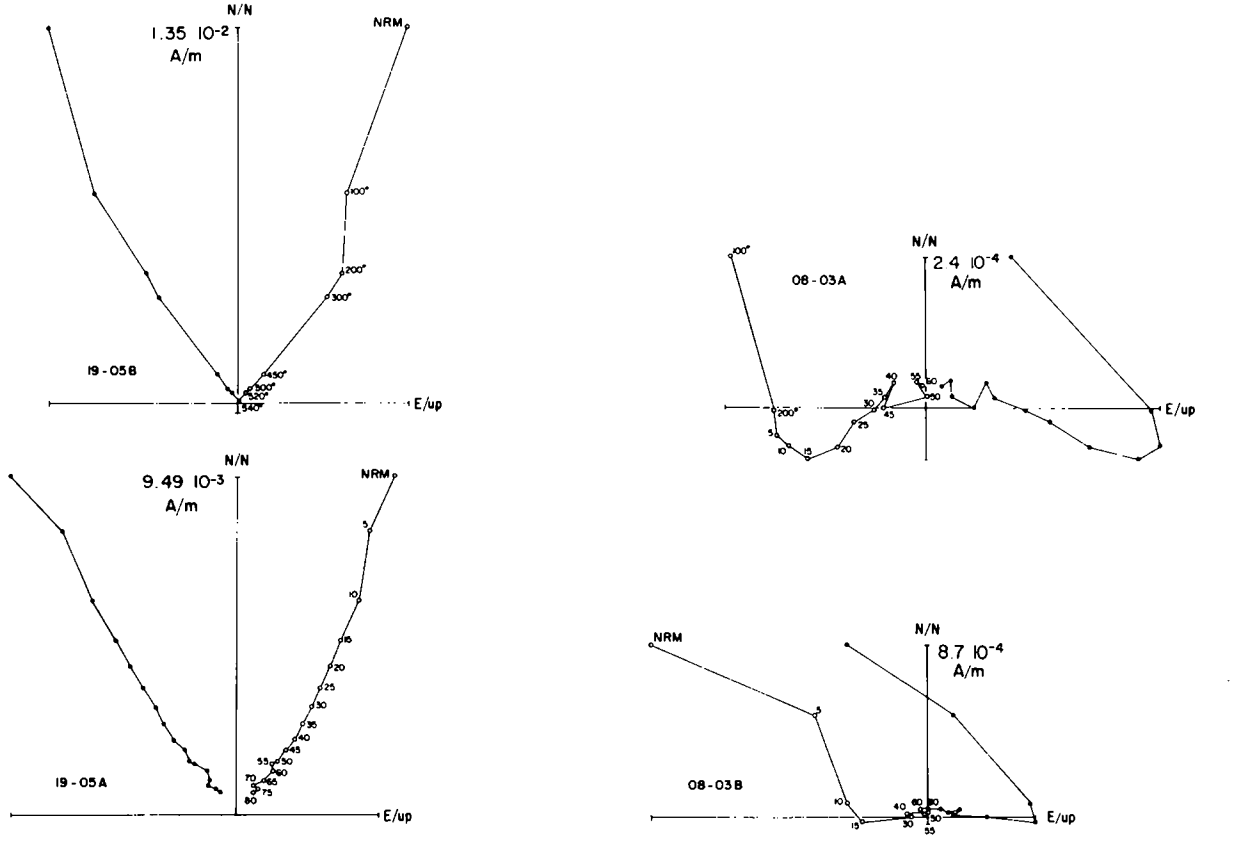


Fig. 3. Orthogonal vector projection of thermal and AF demagnetizations of specimens from the Upper Triassic Huai Hin Lat Formation. Solid symbols: projection on the north-south vertical plane. Open symbols: projection on the horizontal plane. Treatment is indicated near the horizontal projection (in degrees Celsius or millitesla). For sample 08-03, AF demagnetization is compared with mixed thermal and AF treatment.

perimposed with an ambiguous normal component (component A?). This is illustrated by the poor values of the statistical parameters for site 8 (Table 1).

A principal component analysis [31] of the directions showed component A to be parallel to the present field direction in geographic coordinates and thus, most likely, a soft secondary component; only component B will be discussed hereafter.

In the conglomerate (site 5), both components A and B are coherent between the pebbles (Fig. 3), whereas thermal treatment only reveals component A (present field) in the matrix. This is consistent with the coarse-grained lithology of the matrix which makes it unlikely to have preserved a stable magnetization over geological time.

Analysis of the results. In only 26 cores could a stable component of magnetization be isolated. This one-to-three rejection percentage is rather high; yet it is smaller than the one observed by Maranate [16] for most of the Khorat Group (54%). Still, Maranate's study led to a consistent magnetostratigraphy [17]. It indicates that, despite the frequent occurrence of alteration or remagnetization, the primary magnetization is preserved in continental deposits of the Khorat Group. Bedding corrections are always very small for most of the Khorat Group. In the Huai Hin Lat Formation the dip generally does not exceed 20°, except in site 8 where it reaches 35°. Folding attitude does not change much over the area of sampling and the fold test (Fig. 4) relies on site 8. The fold

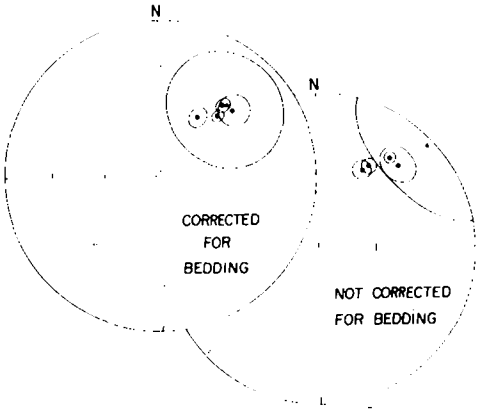


Fig. 4. Equal-area projection of mean site directions with their circles of 95% confidence for the Huai Hin Lat Formation indicating a positive fold test. Antiparallel direction is also indicated (triangle) for site 8 (reversed).

test is positive at the 95% probability level ($K_S/K_G = 4.4$ for 5 sites; Table 1 and Fig. 4) [32]. It leads to a paleomagnetic direction $D = 43°$, $I = 43°$, $\alpha_{95} = 7.5°$, $K_S = 106$. However, the large uncertainty in the result for site 8 warrants further search for evidence that the magnetization has a primary origin. The presence of one reversed direction out of five may provide such evidence, and is not inconsistent with the predominant normal polarity of the Norian geomagnetic field [33]. A stronger evidence is given again by the magnetostratigraphy of the Khorat Group provided by Maranate and Vella [17] which extends down to the Middle Triassic Lom Sak Formation. There is no field evidence that might suggest that primary magnetization has not also survived in the interlayered Huai Hin Lat Formation.

4. Discussion

To the best of our knowledge, there are no other reliable Upper Triassic paleomagnetic results from the Indochina block. The Lom Sak Formation, originally assigned to the Cretaceous by Iwai et al. [34], then shown by Bunopas [35] to be pre-Jurassic, has been tentatively assigned to the Middle Triassic by Maranate [16] and Maranate and Vella [17] on the basis of magnetostratigraphic inference. This age has been recently confirmed by the discovery of vertebrate remains [36]. However, the large uncertainty in the Lom Sak paleomagnetic direction ($\alpha 95 = 14°$ in Maranate [16]; 18° in Bunopas [20]), precludes any geodynamic interpretation. Barr et al. [19] report a Middle Triassic to Lower Jurassic pole which is discussed by Maranate [16]. This pole apparently includes Permian samples but is mostly dominated by Jurassic samples. Although this pole cannot be safely merged with ours, it is not inconsistent with it.

The new Upper Triassic pole derived in this paper can be combined with previously published post Triassic poles for the Indochina block [16,19,20] in order to define what appears to be a rather continuous Mesozoic APWP for this block (Table 2). Changes in paleolatitude and declination, are given in Fig. 5 as a function of time, for a reference point located in the Khorat plateau (16°N, 103°E). We also show for comparison the paleolatitude and declination which would be expected for Indochina, had it remained fixed with respect to Eurasia. These values are computed using Eurasia's APWP from Irving [37] for the Jurassic and older times and the Cretaceous pole from Achache et al. [15]. Also shown on the same diagram are the difference between the Indochina and Eurasia directions.

Three distinct phases of relative motion are apparent in Fig. 5: from Upper Triassic to Middle Jurassic (205–160 Ma), from Middle Jurassic to Upper Cretaceous and from Upper Cretaceous to the present. The most recent phase corresponds to a large clockwise rotation of Indochina with respect to Eurasia with no resolvable relative latitudinal motion. This has been interpreted by Achache et al. [15] as a consequence of the collision of India, thus providing evidence for the extrusion and rotation of Indochina, associated with crustal shortening in south central Asia, as proposed by Peltzer et al. [13] and Tapponnier et al. [7,14]. The tectonic analysis put forward in these papers indicates that most of the rotation is likely to have occurred in the last 40 or 50 Ma as indicated in Fig. 5 (see also Achache et al. [30] Besse et al. [38], and Patriat and Achache [39]. The precise timing of this rotation is, however, not yet determined.

The period from Middle Jurassic to Upper

TABLE 2

Observed versus Eurasian expected paleofield direction, in the Khorat

Age of pole and references	Observed direction for Indochina (Khorat)				Expected direction from Eurasia's APWP (at 16°N, 103°E)				Displacement of Indochina with respect to stable Eurasia	
	D	δD	p	δp	D	δD	p	δp	ΔD (clockwise)	Δp (northward)
Middle Cretaceous (90–100 Ma) Maranate [16]	46.6	14.4	14.0	14.0	17.9	7.2	19.9	6.8	28.7 ± 16.1	5.9 ± 15.6
Upper Jurassic (150–70 Ma) Bunopas [20]; Maranate [16]	29.0	8.2	20.9	7.7	13.6	6.6	24.9	6.0	15.4 ± 10.5	4.0 ± 9.8
Lower Jurassic (170–200 Ma) Barr et al. [19]; Bunopas [20]; Maranate [16]	35.1	5.0	20.7	4.7	24.1	14.8	37.5	11.7	11.0 ± 15.6	16.8 ± 12.6
Upper Triassic (205 Ma) this study	43.2	7.8	24.9	7.0	38.8	4.2	40.0	3.2	4.4 ± 8.8	15.1 ± 7.7

D, p = declination, paleolatitude (directions for Indochina are computed at 16°N, 103°E from VGPs).
δD, δp = errors in declination and paleolatitude. $\delta D = \sin^{-1}(\sin A_{95}/\sin p_0)$; $\delta p = A_{95}$ where A_{95} is the radius of 95% confidence limit of the VGP and p_0 the paleocolatitude.
ΔD, Δp = declination and paleolatitude differences (observed minus expected).
Errors are computed after $X^2 = X_1^2 + X_2^2$.

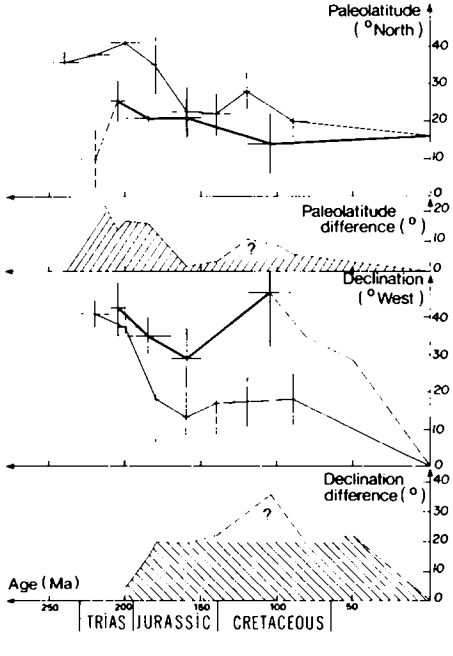


Fig. 5. Comparison of Mesozoic APWP for Indochina (heavy line) and Eurasia (thin line). The upper curves show the comparison of paleolatitudes computed at the point 16°N, 103°E. The dashed line indicates the paleolatitude given by the Middle Triassic Lom Sak Formation. The main changes in paleolatitude occur in Upper Triassic to Lower Jurassic time. The lower curves show the comparison of declinations at the same reference point. Anticlockwise rotation of Indochina with respect to Eurasia is observed in Upper Triassic to lowermost Jurassic time and clockwise rotation sometimes during the Cenozoic. The small differential motion indicated in the Lower Cretaceous relies on a single point from the APWP of Indochina and may not be significant.

Cretaceous appears to be one with little resolvable change, either latitudinal or rotational, between Indochina and Eurasia. Some differential motion is suggested in Fig. 5 during the Cretaceous but relies entirely on the poorly defined Middle Cretaceous pole for Indochina ($\alpha_{95} = 14°$). The Upper Jurassic and the Cretaceous seem to be times of tectonic stability when Indochina, welded to the

South China and Shan-Thai blocks, formed the southern margin of the Eurasian continent as outlined in Achache et al. [15, fig. 5] and Tapponnier et al. [7, fig. 13].

Of particular interest is the older Upper Triassic to Middle Jurassic phase where significant differential motion is indicated by the present data. During this period, Indochina remained at roughly constant tropical northern latitudes. Eurasia on the other hand reversed from a slow northward motion to a fast southward one. As a result, the latitude difference between the two units decreases from $15 \pm 7°$ (2σ level) to a statistically negligible value implying 1650 ± 850 km of latitudinal displacement of Indochina with respect to stable Eurasia. This convergent motion seems to be accompanied by a counterclockwise rotation of Indochina of $15 \pm 10°$.

Two kinds of interpretations can be proposed to account for this convergence: one involves the closure of oceanic domains, the other intracontinental deformation. Since the Eurasian poles used for our comparison are relevant to the Russian platform and to Siberia, the convergence can have been taken up anywhere between Indochina and stable Siberia (which is bounded to the south by the Altai and Yablonov Ranges to the north of China).

The occurrence of a 1000–2000 km wide ocean in the Upper Triassic immediately north of Indochina can apparently be ruled out. It would contradict tectonic evidence for a pre-Upper Triassic closure along the Song Ma suture (see above and Fig. 1) and paleontological observations in the Huai Hin Lat Formation. The discovery of a toothplate belonging to a freshwater ceratodontid closely related to similar continental vertebrate species in South China establishes a continental connection between Indochina and the South China block in the Norian [25]. The 3°N paleolatitude of South China (Yangtze) in the Upper Permian [40] is not inconsistent with a suturing of these two units in Middle Triassic time; it would imply less than 9 cm/yr of absolute velocity between Upper Permian (240 Ma) and Middle Triassic time.

Indochina is thus likely to have been sutured with South China to the northeast and with the Shan-Thai block to the west by Upper Triassic time. The differential motion suggested by the present study should thus have occurred between this continental agglomeration and North China, along the Qin Ling and Bakan Hai Ranges (or Songpan-Ganzi—see Fig. 1). Indeed, a preliminary Middle Triassic pole for South China [41] indicates no significant latitudinal motion with respect to Indochina since that time. Furthermore, Lin's [41] paleomagnetic study reveals significant Lower and Middle Jurassic latitudinal convergence between South China and North China. Geological data on the Songpan-Ganzi are still scarce. Thick series of marine deposits are observed: Permian deep-sea sediments are overlain by Triassic flysch of predominantly Tethyan affinities, intercalated with volcanics. Facies become gradually shallower upwards and marine sedimentation seems to terminate in the latest Triassic [42]. This may indicate the closure of an ocean basin which would account for part of the observed motion. However, the lack of Jurassic sediments suggests that the final closure occurred prior to the Jurassic and thus it could only account for the initial part of the motion.

A second interpretation involves intracontinental deformation. Paleontological observations in the Huai Hin Lat also reveal several continental species which exhibit remarkable affinities with Eurasian vertebrates (stegocephalians, turtles, phytosaurs) and imply the existence of continental connections between Indochina and central Eurasia prior to the Upper Triassic. Thus, the welding of South China and Indochina onto Eurasia must have begun prior to or during the Upper Triassic. After this initial collision, relative motion would have continued until Upper Middle Jurassic, resulting in continental shortening in southeastern Asia. This crustal shortening could be related to deformation and mountain building to the north of the South China block and as far north as the North China block (see, e.g., Klimetz, [43]). Indeed, a preliminary Middle Triassic pole for South China [41] indicates some 20° of clockwise rotation of this block with respect to Indochina and Jurassic poles obtained in the same study [41] from different places in North China indicate important internal deformation within the North

China block (see Achache [44]). Also, a paleomagnetic study of Upper Permian formations [40] places North China (or Sino-Korea) near 10°N in the uppermost Permian, i.e., more than 50° to the south of Siberia. The absence of latitudinal motion between these two units in post-Norian time would imply an unlikely high minimum mean relative velocity of 18 cm/yr of the North China plate with respect to Siberia during the Lower Triassic.

The observed convergence would also be consistent with the mechanism proposed by Sengör and Hsü [6], who interpret the Songpan-Ganzi system as a giant flysch/mélange-filled accretionary complex. They believe that the Paleotethys had evaporated by Upper Triassic, allowing the migration of continental species, but that convergence continued by "hidden subduction" of the oceanic lithosphere underlying the accretionary complex until final continent to continent collision occurred in the Early Cretaceous. Available field evidence and geophysical data are, however, not sufficient to further test this "hidden subduction" model.

5. Conclusion

The synthesis of paleomagnetic and paleontological data from the Khorat Group indicates that Indochina was welded to Eurasia by the upper Middle Jurassic. The initial contact between the two continental domains occurred prior to the Upper Triassic and resulted in the partial closing of the intervening Paleotethys Ocean. Thus, the process of accretion of Indochina against the southern margin of Eurasia seems to have lasted from the Upper Triassic to the late Middle Jurassic. This process occurred at quasi-constant northern tropical latitudes .

Relative motion between Indochina and Eurasia resulted in several hundred kilometers of convergence (1650 ± 850 km). Two mechanisms may account for this convergence. This first one involves the closure of intervening oceanic areas (remnants of the Paleotethys) between South China and Eurasia. The second one involves actual shortening of the continental crust in southeastern Asia. However, more paleomagnetic data of Triassic and Jurassic age in mainland southeastern Asia are needed to confirm the observed motion and further test the different models proposed in this study.

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